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A Transdisciplinary Framework for Handling Disruptions in Vehicle Routing with Customer Time Windows

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Abstract. The vehicle routing problem (VRP) consists of servicing several locations within a period under some requirements (customer time windows, work shifts, etc.). Unforeseen changes in those requirements during plan execution, alias disruptions, are common in practice. They pose significant challenges to logistics operations due to real-world circumstances' increasing unpredictability and delayed agility of the business. Stakeholders, such as customers, vehicle crews, company executives, or other third parties, are impacted qualitatively and quantitatively. Extensive research has been dedicated to VRP under disruptions, such as stochastic VRP (planned incorporation of uncertainty) and dynamic VRP (real-time data changes during a running operation). However, academic literature on the latter involving multiple stakeholders is scant. This is the motivation for this work, which involves both management disciplines (like accounting, finance, marketing, and sustainability) and engineering fields (such as logistics, mathematical programming, and computer science). The contribution of this article is twofold: first, a transdisciplinary framework for addressing stakeholders and environmental needs for dynamic vehicle routing disruption management is proposed. Second, an industrial-size illustrative instance is solved to show the framework's benefits. Deviations from original plans tend to be the rule in many industries and significantly affect operations' carbon footprint. In addition, they affect businesses' sustainability (economic, societal, and environmental). Consequently, this approach might measurably mitigate negative impacts resulting from operational disruptions while pioneering transdisciplinarity in logistics.

Keywords. Disruption management, social sustainability, stakeholders, transdisciplinary engineering, vehicle routing.

Introduction

Traffic congestion is a growing challenge in urban areas, compounded by environmental stress from emissions and triggering road accidents - the financial impact alone costs the EU 1.4% of its GDP and the US 0.7%, with conditions worsening in Latin America due to rapidly rising motorisation of the population [1] and will therefore stay an important challenge in the future. Those deviations significantly impact logistics operations, making it difficult to maintain efficient vehicle routing via urban road networks.

The origin of linear programming representations of routing models by Dantzig [2] was soon to be generalised to more detail, one milestone marked by Solomon modelling multiple vehicles and time-windows (VRPTW) [3]. Dynamic VRPs, i.e. VRP under

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changing parameters during execution, have been examined with different causes of the dynamism [4, 5]. Disruptions differ from minor deviations in that they require significant operational changes [6]. In the case of a VRPTW, this would be the infringement of customer time windows or exceeding the scheduled shift length and leads to the disrupted VRPTW (DVRPTW). This DVRPTW and its stakeholder implications are the core problem examined in this paper.

While governments are primarily responsible for infrastructure management, businesses with logistics operations play a crucial role in mitigating these challenges through strategic decision-making. Stakeholders affected include logistics companies, vehicle drivers, customers, warehouse staff, and operations planners. Additionally, other road users, regulatory bodies, worker unions, and non-profits are indirectly influenced by inefficiencies in logistics networks.

The inability to manage disruptions effectively can lead to economic losses, delays, and negative environmental impacts. Existing research has explored solutions to loosely related problems but never dealt with this application. However, most approaches focus on cost minimisation rather than a holistic, stakeholder-driven response. Merely extreme scenarios, like natural disasters [7] or even war zones [8] differ quite starkly from classic VRP approaches, i.e. equipment protection and human wellbeing are emphasised over pure financial objectives [8, 9].

Common strategies to tackle expected plan deviations include:

1. Dynamic Programming: the entire system is recalculated as soon as instance changes become known [4], whereas approximate dynamic programming [10] is needed for large instances.
2. Stochastic VRP: some discrete model elements are exchanged for stochastic values or random variables [4, 11, 12].
3. Robust VRP: due to intentional buffer integration, the solution is therefore less optimal but less sensitive to parameter changes [4, 12, 13].

All three approaches trade off solution optimality for expected deviations. Approaches differ between maximising demand fulfilment on the long run (while accepting unsatisfied demand) [14] or mandatory full recovery of missed actions [6]. An example for mitigation strategies for recovery scenarios respecting stakeholders is presented in [15]. The authors of [16] used a multi-objective objective to model flight trajectories incorporating delays, flight efficiency and route charges while, therefore balancing different financial impacts. Even legislative restrictions and their trade-offs with pure cost efficiency have been evaluated [17].

Therefore, most of the current literature focuses on the company's financial point of view only but this approach is incomplete. There are more stakeholders to take care of along with qualitative measures to consider. To close this gap, research should consider zooming out and holistically include a broader range of stakeholders into the problem-solving process.

Holding the potential to facilitate this type of problem-solving, the transdisciplinary research approach comes into play: [18] yield a comprehensive insight to the patent landscape from a blend of engineering and legal expertise. The creation of vibrant entrepreneurial ecosystems is the objective of [19], which leverage a mix of engineering, social sciences, systems theory and regional development research. A demonstration of how electrical engineering may be applied to finance and business topics can be seen in [20], where the performance of new ventures is predicted. Elements from reliability engineering were shown to be suitable methods to better understand failures in education [21]. Subjective preferences of students and professors can improve the success of professor-course assignments in practice [22]. An extension of the VRP to the United Nation's Sustainable Development Goals was achieved by [23], introducing consumer-driven factors and the selection of low-impact delivery windows. [24] show how Transdisciplinarity (TD) benefits all stakeholders of routing problems by combining social sciences and mathematical rigor.

To tackle this rather complex issue, a Transdisciplinary DVRPTW (TD-DVRPTW) is proposed. All methodological steps also fit to disrupted VRP in general (TD-DVRP). This work combines operations research, data science, traffic engineering, psychology, and economic modelling to manage disruptions in vehicle routing.

The contributions of this paper are therefore firstly, the development of a structured, stakeholder-inclusive disruption management framework for vehicle routing with time windows (VRPTW) and secondly, a numerical case study demonstrating its practical benefits.

Preliminary results show that the framework enhances adaptability in logistics operations by balancing efficiency with stakeholder needs. This work directly aligns with the idea to make the world a better place, illustrating how interdisciplinary collaboration improves real-world problem-solving in dynamic and uncertain environments. Especially the potential spill-over effects of the solution from primary decision-makers and stakeholders to other users of the road network showcase alignment with public welfare improvement.

1. Methodology and proposed transdisciplinary framework

The TD framework for integrating stakeholder perspectives into the DVRPTW is summarised in Table 1. It merges elements from related methodologies (derived from extensive literature review) and was refined iteratively through multiple instances [25]. It emphasises stakeholder impact evaluation and encourages a cyclical approach to continuously update data collection efforts.

Table 1. Structured Transdisciplinary Framework for Disruption Management in VRP (TD-DVRP)

Phase	Key Actions
I: Stakeholder Engagement and Planning	1. Stakeholder identification (internal/external, private/public) 2. Transdisciplinary steering: clarification of common objectives and concerns (e.g., feasibility, satisfaction, finance, environment, reputation, control) 3. Formation of a transdisciplinary working group
II: Structural Data Building and Disruption Modelling	1. Define operational parameters (fleet size, shift duration, customer locations, depots, travel cost matrix) 2. Define data granularity and select collection methods. 3. Incorporate disruption management policies: a) Identify possible mitigation measures. b) Evaluate each measure's impact on stakeholders. 4. Data calibration using historical disruption and stakeholder insights (after looping up from phase III)
III: Disruption Resolution and Transdisciplinary Decision-Making	1. Develop a mathematical VRP & and TD-DVRP model. 2. Define disruption-causing deviations. 3. Model implementation and solving. 4. Conduct stakeholder-focused sensitivity analysis. 5. Model calibration and feedback loop to Phase II (in practice)

1.1. MIP for VRPTW

For the base case multiple-depot VRPTW, a mixed-integer formulation with the following properties is necessary: minimise total travel cost, each vehicle departs and returns to the depot once, each customer is visited once, routes are continuous and cannot create isolated loops, vehicles must arrive within time windows.

Solving the VRPTW requires distance or travel time matrices, with customer addresses converted to geocoordinates using various tools [26, 27, 28]. Understanding customer sensitivity to violations depends on their operations and context.

The solution determines which arcs are used and when customers are served, this information can be transformed to form routes for each vehicle. Scalability is a challenge, adding just two customers to a 100-customer problem increases complexity over 10,000 times. Uncertain disruptions further limit reaction time, complicating real-time decision-making.

1.2. From plan dynamism, deviations to disruptions

After deriving the operational plan from the VRPTW solution, vehicles' travel times may deviate due to unforeseen events. The recalculation starts with a "cut-off" version of the original instance at disruption time (t_D), as vehicles may already have served customers. The later the cut, the smaller the instance, reducing computational complexity depending on the customer order.

1.3. Disruption modelling and mitigation

In some cases, typically when only slight deviations occur, the initial VRPTW can simply be recalculated or stays feasible due to buffers. If the VRPTW has no feasible solution, a DVRPTW must be formulated, including suitable mitigation measures.

A mixed-integer formulation for the disrupted, transdisciplinary VRPTW includes the following properties: minimise the travel costs and negative stakeholder impacts (minimise overtime for drivers, evaluate utilisation of backup trucks, minimise outsourced customers, minimise the violation of time windows), each vehicle departs and returns to the depot, each customer is visited exactly once or skipped for a penalty, routes are continuous, i.e. no isolated loops between the nodes can be formed, each customer must be serviced within his respective time window or a penalty applies, time window violations are the total amount of chosen service times outside of the customer's time window boundaries, definition of overtime.

Effective communication of new solutions to drivers is crucial. While stepwise updates simplify frequent changes, psychology suggests full-shift waypoint disclosure improves engagement [29]. However, security and data protection must guide data availability adjustments [30].

1.4. Solution steps

When dealing with large instances and especially when limited to demanding reaction times, heuristic and metaheuristics solvers are the method of choice. A broad range of those algorithms have been developed and applied to different VRP variations [31, 32]. It is assumed that for solving the initial VRPTW much more solver time is available than for the DVRPTW, because the VRPTW is solved in-between shifts while the DVRPTW is about quickly re-solving the disrupted instances during running operations. To simulate this discrepancy, the VRPTW receives one hour solver time and the DVRPTW one minute – both values are chosen based on experimental practice [25].

Two algorithms are used to initialise and improve the solution:

1. Initial solution using Path Cheapest Arc [33] algorithm: a heuristic for VRPs, like the more well-known metaheuristic Nearest Neighbour algorithm. This greedy method simply builds a route connecting cheapest nodes without minimising the global distance.
2. Then, the initial solution is further optimised using a Guided Local Search algorithm which is a machine learning enhanced heuristic [34]. It performs exceptionally well on routing problems [33, 35].

1.5. Modelling qualitative stakeholder input

Qualitative stakeholder input was generated using the novel economic modelling technique of *Homo Silicus*, which is empirically proven to "qualitatively recover findings from experiments with actual humans" [36]. It can be seen as an extension of the well-known Homo Economicus model but capable of more individual subtleties to its decisions [36]. For each identified stakeholder, a Large Language Model (LLM) instance [37] is (context-) prompted to behave like a member of the transdisciplinary team and reply from the viewpoint of its respective stakeholder interest. Each resulting agent i (of the set of transdisciplinary team agents τ) is then prompted to express a severity value l_{ip} to each mitigation policy p (of the set of possible policies P), ranging from 1 (acceptance), 5 (abstention) up to 10 (rejection).

$$\boldsymbol{\lambda_p} = \frac{\left(\boldsymbol{\Pi}_{i=1}^{|\tau|} \boldsymbol{l_{ip}}\right)^{\frac{1}{|\tau|}}}{\mathbf{5}} \forall \boldsymbol{p} \in \mathbf{P}, \mathbf{0.2} \leq \boldsymbol{\lambda^p} \leq \mathbf{2} \tag{1}$$

Using a Equation 1, the survey results are converted into a weighted stakeholder preference rating (scalars λ^p), which will modify the cost function of the model and thereby influence its decision-making [25].

Limitations of LLMs like possible bias or hallucination[2] do not interfere with the proposed method, since the framework does not claim to provide the best mitigation policy itself. It merely shows how to quantify the qualitative feedback to adjust the decision on chosen mitigation policies, independent of the feedback details. In a fast and scalable manner, the LLM results provide a first glimpse how the decision-making of the involved stakeholder will differ before conducting more sophisticated investigation.

1.6. Numerical illustration

In this section, the proposed framework is applied step by step to a real-word business use case. Subject of the investigation is the road distribution network of a company in a Latin-American metropolitan area, serving 280 customers each shift .

Step I-1: Company internal stakeholders are vehicle drivers, the company owner (or general management), operations and logistics manager, sustainability representative and

[2] The interested reader may refer to [36].

risk manager. External ones are primarily the customers, the customer's employees and NGO representatives.

Step I-2: General objectives for the transdisciplinary research process are operational feasibility, timely termination, customer satisfaction, financial implications, environmental impact, the company reputation and controllability of operations.

Step I-3: Instead of assembling an entire team of transdisciplinary experts, the stakeholders' roles are represented by the economic agents according to the *Homo Silicus* model. This allows for quick, approximate results which are sufficiently different to demonstrate the model functionality.

Table 2. Merging financial costs and stakeholder input. For each measure (col. a) the financial cost (col. b) is multiplied by the stakeholder preference λ (col. c) to generate the final scalars (col. d) which tweak the model solution outcomes by modifying the objective function. The higher that value in col. d, the the less likely the model will use the respective measure to recover the disrupted VRP instance.

a) Identified disruption mitigation measures from Step II-3 (p)	b) Financial Cost of measures (monetary units)	c) Stakeholder Preference (λ_p)	d) Cost factor in $\lambda_p{}^2$-TD-DVRPTW [cost*$(\lambda_p)^2$]
Overtime for Drivers	2 /min	1.03 (neutral)	*2.18 /min*
Utilisation of Backup Trucks	400 /vehicle	0.62 (preference)	*154.75 /vehicle*
Outsourcing Deliveries	50 /customer	1.66 (avoidance)	*138.46 /customer*
Early Service	2 /min	0.88 (preference)	*1.54 /min*
Late Service	3 /min	1.72 (avoidance)	*8.83 /min*

Step II-1: The travel time matrix and customer time windows are fetched from [24] and are available online.[3]

Step II-2: Since the data already given, no evaluation of data granularity and collection methods is needed. An elaborative example of this step is demonstrated in [25].

Step II-3: Mitigation strategies for disruption which aim to re-establish solution feasibility in the case of severe disruption while respecting the common goals are overtime for drivers, outsourcing deliveries, utilising additional vehicles from the resting fleet, early service of customers and late service of customers. Table 2 shows how the transdisciplinary agents evaluated the stakeholder impact using the methods described in section 1.5. The computations were executed on a local PC.[4]

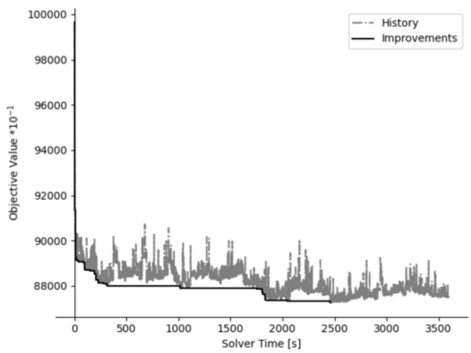

Figure 1. The initial VRPTW objective value vs. solution time. Even after 1500s, visible significant improvements are made.

Step II-4: Data calibration cannot be done at that point, since no historical data is available. A sample execution of this step can be found in [25].

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[3] https://www.dropbox.com/scl/fi/ogkohllsyfcz2lrbhd43t/Paper-TE2024-VRPTW-Data-and-Sol.xlsm?rlkey=2w1k38pajfxaria8zc8a4ub70&dl=0 and https://github.com/mx-bt/paper-td-dvrptw.git

[4] Operating System: System Type: x64-based PC; Processor: Intel(R) Core(TM) i7-10700F CPU @ 2.90GHz, 2901 MHz, 8 Cores, 16 Logical Processors; Graphics Card: NVIDIA GeForce RTX 2060; Physical Memory: (RAM) 16.0 GB

Step III-1 & -2: Two mixed-integer-programming models, the definition of a disruption for this context and all necessary data transformations are outlined in sections 1.1-1.3.

Step III-3: The mathematical model is implemented in Python code to access heuristic solvers for the solution steps described in 1.4, using the Google OR-Tools library [33]. The initial solution of the instance was reached after ~40min solver time as seen in Figure 1: the customers are served by 26 routes reaching an objective value[5] (OV) of 8722.8 with a total time of all routes[6] 12073 min. Solution details can be found on GitHub.[7]

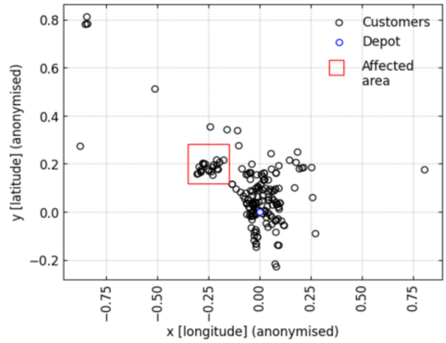

Figure 2. Visualisation instance nodes (customers and depot) and congested area. At t_D=240min, all in- and outcoming routes to the customers in the red area are affected by a travel time increase by the factor of 3.2 [1].

As an example, one may imagine a political demonstration, taking place within the distribution network, as described in Figure 2. Transferred to the TD-DVRPTW cost function, this deviated solution would have a strongly worsened OV of 29274.7– more than four times higher than the initial OV – and delayed total time of 13361.9 minutes. Both measures indicate severe negative impact financially and regarding stakeholder satisfaction.

After 60 seconds of λ^2-TD-DVRPTW re-optimisation (see Figure 3) an OV of 7978.7 was reached by dropping 7 customers (most of them within the affected area) and mobilising 4 vehicles from the resting fleet.

Step III-4: In this case, different weightings of the quantified stakeholder inputs were tested as illustrated in Table 3. Scaling up the λ-Values leads to solutions closer to the stakeholder preferences like avoiding customer drops and incorporating more vehicles of the own fleet as well as the early service of customers. A behaviour which clearly reflects the stakeholder votes from Table 2.

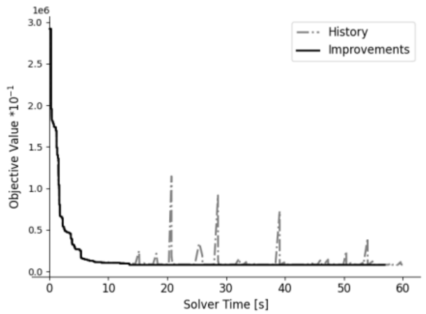

Figure 3. Solution progress relative to solver time. Since a good solution is available after ~20 seconds already, the model implementation seems suitable for realistic instances.

Table 3 shows the change of chosen mitigation measures as the power of λ increases. The model is highly sensitive to the chosen scalars λ of the objective function – adapting the chosen measures according to stakeholder inputs.

The proposed framework allowed for a representation of subjective stakeholder interests in the mathematical models decision-making by assigning a weighting to each possible mitigation policy in the objective function. It also showed how severe a plan deviation (here: traffic congestion) would worsen stakeholder satisfaction and how the re-solving successfully mitigates this issue.

[5] Corresponds to vehicle driving minutes * 1 monetary unit operating cost.
[6] Corresponds to the sum of vehicles' arrival times, including waiting times.
[7] https://github.com/mx-bt/paper-td-dvrptw.git

Table 3. Sensitivity analysis on the different weightings of stakeholder interests in the cost function.

Mitigation Measure In TD-DVRPTW (p)	λ_p	$(\lambda_p)^2$	$(\lambda_p)^3$	$(\lambda_p)^5$
Overtime for Drivers	0	0	*0*	*0*
Utilisation of Backup Trucks	0	4	*6*	*7*
Outsourcing Deliveries	16	7	*2*	*1*
Early Service	0	0	*0*	*3*
Late Service	0	0	*0*	*0*

2. Conclusions and further work

This paper introduced a transdisciplinary framework for managing disruptions in vehicle routing with customer time windows (TD-DVRPTW), integrating qualitative stakeholder perspectives into quantitative modelling. By combining operations research, data science, psychology, and economic modelling, we demonstrated that incorporating stakeholder preferences into optimisation models improves decision-making in dynamic and uncertain environments.

The numerical case study illustrated the framework's practical benefits, showing that solutions adapted to stakeholder needs enhance operational feasibility while balancing economic, environmental, and social objectives. Sensitivity analysis further confirmed the impact of weighting stakeholder preferences in optimising disruption response strategies, reinforcing the value of proactive, structured, and holistic disruption management.

At its core, the proposed framework and applied methodology allow not only allows to incorporate stakeholder-individual feedback into mathematical modelling but also provides simulation techniques to estimate qualitative feedback and first sensitivity analysis on the model at hand. The numerical evidence underlines this finding, where the capability of the model to sensitively represent changes stakeholder interests or desired impacts on the outcome is shown.

Route adaptations primarily affect the company and its stakeholders, but a positive side effect is that the TD-DVRPTW re-routing potentially helps vehicles avoid congested areas, improving overall congestion occurrence and contributing to a better traffic situation in general.

The financial scalars used for disruption measures are a simplification; modelling factors like customer dissatisfaction or overtime-induced fluctuation requires more complexity. Improved modelling of these factors could significantly change results.
Further research could explore the trade-off between solver time and system delay—how long to run the solver while accepting vehicle halts? Predictive simulations could monitor disruption sources and start recalculating before issues arise. The unpredictability of congestion and disruptions raises questions about computational capacity. While frequent calculations are best on dedicated hardware, ad-hoc recalculations may need scalable cloud services.

Although combining heuristics is not new, early experiments suggest that passing an instance between multiple heuristics can maintain high improvement velocity. This could lay the foundation for a machine learning approach to optimize solver selection.

Additionally, research could expand to multi-period VRPs and include more complex acceptance rules. It could also consider a wider range of stakeholders, such as the families of drivers, where cultural and psychological factors influence overtime willingness.

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